

New born screening:

- Some genetic disorders can be detected at birth by simple biochemical tests. E.g. screening for DMD by measuring creatinekinase level in newborns.
- Newborn screening is an effective public health strategy for treatable disorders such as PKU, sickle cell disease, hypothyroidism and galactosemia.

Heterozygote screening:

This is can be applied to the detection of unaffected carriers of disease-causing mutations.

Genetic diseases amenable to heterozygote screening are typically autosomal recessive disorders (ex: Tay-Sachs disease, B-Thalassemia ,and cystic fibrosis).

Accurate carrier testing are available for mutations in a target population ,then they can be given information about risk and reproductive options.

As a result of screening, the prevalence of newborn with such disorders are dramatically declined.

Presymtomatic diagnosis:

Individual who are known to be at risk for a disorder can be tested to determine whether they have inherited a disease – causing mutation before they develop clinical symptoms of the disorder. Presymptomatic diagnosis is available, for example, for Huntington disease, adult polycystic kidney disease, hemochromastosis, and autosomal dominant breast cancer. By informing individuals that they do or do not carry a disease- causing mutation, presymptomatic diagnosis can aid in making reproductive decisions.

Direct mutation analysis:

Sometimes the mutation causing a disease happens to alter a recognition sequence for a restriction enzyme. In this case the mutation itself creates a restriction site polymorphism that can be detected after digestion with this enzyme.

An example is sickle cell disease mutation, which alters an *Mst* II recognition site in the B-globin gene.

New methods of direct mutation detection including the use of Microarray, and mass spectrometry, promise to increase the speed and accuracy of genetic diagnosis.

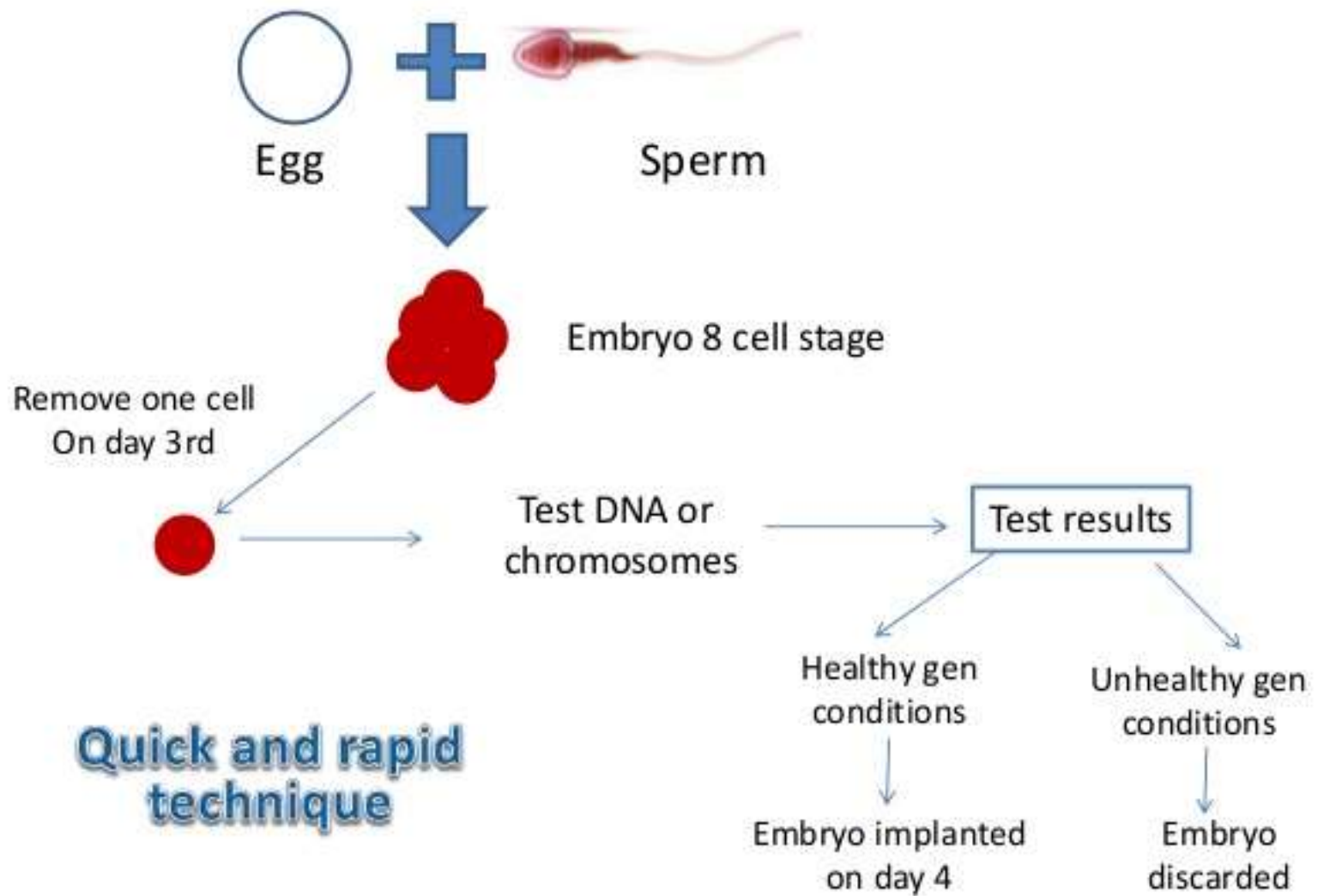
Preimplantation genetic diagnosis

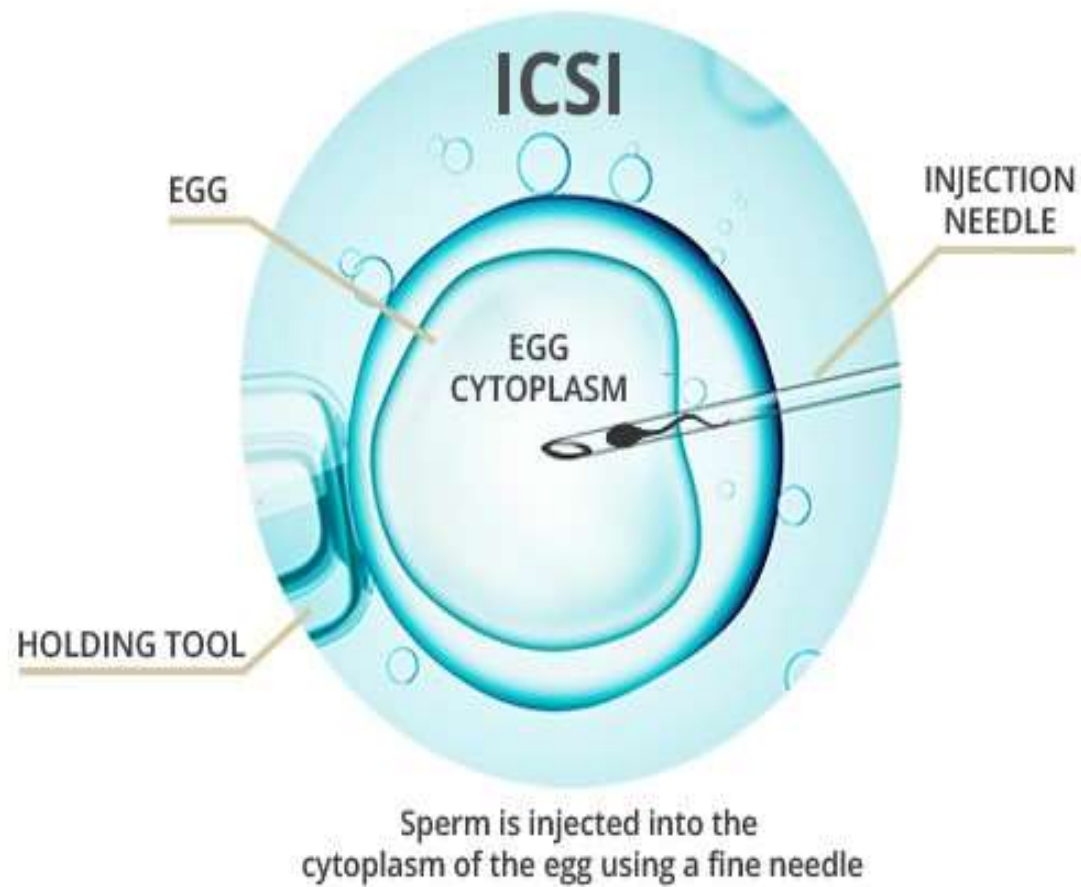
- PGD (also known as embryo screening) refers to procedures that are performed on embryos prior to implantation.
- This technique allows people who carry a genetic defect to avoid producing a child with disorder.
- The procedure begins with in vitro fertilization (IVF) in which hormones (gonadotrophin-releasing hormone and recombinant follicle stimulating hormone) are used to induce ovulation.

- The ovulated eggs are surgically removed from the surface of ovary, placed in a laboratory dish, and fertilized with sperms.
- The zygotes are allowed to divide several times until they reach the 8- or 16-cell stage ,at this stage one cell is removed from each embryo and later tested for the genetic abnormalities

- RT-PCR used to amplifying a minute quantities of RNA transcripts of a single cell) which make it possible to assay for the presence of specific genetic change by(DNA chips, FISH test).
- A healthy desired embryo is selected and implanted in the woman's uterus.

The PGD process





Intracytoplasmic sperm injection